

Burak Ekici, Ph.D.

Senior Research Associate at Department of Computer Science
University of Oxford

BUREAU: 404, Wolfson Building
EMAIL: burak.ekici@cs.ox.ac.uk
URL: <http://ekiciburak.github.io/>
LANGUAGES: Turkish (native), English (operational), French (intermediate),
Italian (intermediate), German (notions)

WORK EXPERIENCE

2023-NOW	Senior Research Associate – Department of Computer Science University of Oxford – Oxford, UK Topic: Multiparty Session Types Advisor: Nobuko Yoshida
2022-NOW	Assistant Professor – Department of Computer Engineering Mugla Sıtkı Koçman University – Mugla, Turkey
2021-2022	Assistant Professor – Department of Computer Engineering TED University – Ankara, Turkey
2019-2020	Assistant Professor – Department of Computer Engineering Istanbul Kültür University – Istanbul, Turkey
2018-2019	Postdoctoral Researcher – Department of Computer Science University of Innsbruck, Innsbruck, Austria Topic: CoqHammer Advisor: Cezary Kaliszyk
2016-2017	Postdoctoral Researcher – Department of Computer Science University of Iowa, Iowa, USA Topic: CVC4, SMTCoq Advisor: Cesare Tinelli
2010-2011	Trainee Researcher – European Commission Joint Research Centre Ispra(VA), Italy Topic: Biometric Security Systems Advisor: Günter Schumacher

EDUCATION

2013-2015	PhD Degree in COMPUTER SCIENCE AND MATHEMATICS <i>Programming Language Semantics</i> , Université Grenoble Alpes – Grenoble, France Advisors: Jean-Guillaume Dumas , Dominique Duval , Damien Pous
2009-2012	MSc Degree in COMPUTER ENGINEERING <i>Parallelization in Functional Programming</i> , Yasar University – Izmir, Turkey Advisor: Ahmet Hasan Koltuksuz
2004-2009	BSc Degree in COMPUTER ENGINEERING Izmir Institute of Technology – Izmir, Turkey Advisor: Ahmet Hasan Koltuksuz

RESEARCH INTERESTS

Computational effects, categorical logic, type theory, Coq, functional programming, SMT, hammer systems, multiparty session types.

PUBLICATIONS AND ABSTRACTS

Journals

- 2026 [Formalising Asynchronous Session Subtyping](#)
ACM Transactions on Computational Logic
Burak Ekici and Nobuko Yoshida.
- 2023 [A Sound Definitional Interpreter for a Simply Typed Functional Language](#)
Axioms
Burak Ekici.
- 2022 [Formal categorical reasoning](#)
Turkish Journal of Mathematics
Burak Ekici.
- 2020 [Mac Lane's Comparison Theorem for the Kleisli Construction Formalized in Coq](#)
Mathematics in Computer Science
Burak Ekici and Cezary Kaliszyk.
- 2018 [IMP with exceptions over decorated logic](#)
Discrete Mathematics and Theoretical Computer Science
Burak Ekici.

Conferences

- 2025 [Formalising Subject Reduction and Progress for Multiparty Session Processes](#)
Sixteenth Conference on Interactive Theorem Proving (ITP – to appear)
Burak Ekici, Tadayoshi Kamegai, and Nobuko Yoshida.
- 2024 [Completeness of Asynchronous Session Tree Subtyping in Coq](#)
Fifteenth Conference on Interactive Theorem Proving (ITP)
Burak Ekici, and Nobuko Yoshida.
- 2023 [Formal Verification of Bit-vector Invertibility Conditions in Coq](#)
The 14th International Symposium on Frontiers of Combining System (FroCoS)
Burak Ekici, Arjun Viswanathan, Yoni Zohar, Cesare Tinelli, and Clark Barrett.
- 2019 [Verifying Bit-vector Invertibility Conditions in Coq \(Extended Abstract\)](#)
Sixth Workshop on Proof eXchange for Theorem Proving (PxTP)
Natal, Brazil, August 2019
Burak Ekici, Arjun Viswanathan, Yoni Zohar, Clark Barrett, and Cesare Tinelli.
- 2018 [Concrete Semantics with Coq and CoqHammer](#)
Proc. 11th Int. Conf. on Intelligent Computer Mathematics (CICM)
Hagenberg, Austria, August 2018
Lukasz Czajka, *Burak Ekici*, Cezary Kaliszyk.
- 2018 [Towards Mac Lane's Comparison Theorem for the \(co\)Kleisli Construction in Coq](#)
Proc. of the 3rd Formal Mathematics for Mathematicians (FMM)
Hagenberg, Austria, August 2018
Burak Ekici.
- 2017 [SMTCoq: A plug-in for integrating SMT solvers into Coq](#)

- Proc. 29th Int. Conf. on Computer Aided Verification (CAV)
Heidelberg, Germany, July 2017
Burak Ekici, Alain Mebsout, Cesare Tinelli,
Chantal Keller, Guy Katz, Andrew Reynolds and Clark Barrett.
- 2016 [Extending SMTCoq, a Certified Checker for SMT \(Extended Abstract\)](#)
HaTT 2016: 1st International Workshop on Hammers for Type Theories
Coimbra, Portugal, July 2016.
Burak Ekici, Guy Katz, Chantal Keller, Alain Mebsout,
Andrew Reynolds and Cesare Tinelli.
- 2015 [Hilbert-Post completeness for the state and the exception effects](#)
MACIS 2015: Sixth International Conference on Mathematical Aspects
of Computer and Information Sciences, Berlin, Germany, 11–13 November 2015.
Jean-Guillaume Dumas, Dominique Duval, *Burak Ekici*,
Damien Pous and Jean-Claude Reynaud.
- 2014 [Certified proofs in programs involving exceptions](#)
CICM 2014 : Eighth Conference on Intelligent Computer Mathematics
Coimbra, Portugal, 7–11 July 2014, CEUR Workshop Proceedings, no 1186, paper 20.
Jean-Guillaume Dumas, Dominique Duval, *Burak Ekici* and Jean-Claude Reynaud.
- 2014 [Formal verification in Coq of program properties involving the global state effect](#)
JFLA 2014 : Journées Francophones des Langages Applicatifs, Fréjus,
France, 8–11 January 2014
Jean-Guillaume Dumas, Dominique Duval, *Burak Ekici* and Damien Pous
- 2010 [Implementations of Category Theory in Computer Science](#)
9th National Mathematics Symposium, Karadeniz Technical University, October 2010.
Burak Ekici, Cagatay Yücel, Gökem Kılınç, Ahmet H. Koltuksuz.
- 2010 [Implementation of Search Strategies in Control Network Programming](#)
Procedural and Non-Procedural In: Proc. Intl. Symposium in Innovations in
Intelligent Systems and Applications (INISTA 2010),
June 2010, Kayseri, Turkey, 386-390.
K. Kratchanov, E. Golemanova, T. Golemanov, T. Ercan, *B. Ekici*.

Abstracts

- 2022 [Formal Categorical Reasoning](#),
Studies on Scientific Developments in Geometry, Algebra, and Applied Mathematics,
(Online Event) 1-3 February 2022.
Burak Ekici.
- 2014 [Program certification with computational effects](#),
Journées Nationales de Calcul Formel (JNCF) 2014, Marseille, France November 2014.
Jean-Guillaume Dumas, Dominique Duval, *Burak Ekici* and Damien Pous.
- 2014 [Certification of programs with computational effects](#),
Doctoral session. CICM 2014 : Eighth Conference on Intelligent Computer Mathematics,
Coimbra, Portugal, 7–11 July 2014.
Burak Ekici.

TEACHING

2023-2024	Invited Lecturer	Distributed Processes, Types and Programming
2023-2024	Invited Lecturer	Introduction to Proof Systems

2022-2023	Main Course	Operating Systems (System programming in C)
2019-2020		
2022-2023	Main Course	Programming Language Concepts (OCaml, Coq)
2020-2023	Main Course	Formal Languages and Abstract Machines
2022-2023	Main Course	Functional Programming (Haskell, Coq)
2021-2022	Main Course	Category Theory in Computer Science (Coq)
2021-2022	Main Course	Fundamentals of Programming I (Java)
2019-2020	Main Course	Digital Design
2019-2020	Main Course	Microprocessors (x86)
2018-2019	Proseminar	Discrete Mathematics

ADVISEES

Omer Keskin (MSc Thesis at CS Department) University of Oxford
 Formalising a strong normalisation result for typed π -calculus in Coq.
 supervisor: Nobuko Yoshida

Quanwen Chen (MSc Thesis at CS Department) University of Oxford
 Formalising Mac Lane's comparison theorem for the Eilenberg-Moore constructions in Coq.
 supervisor: Nobuko Yoshida

Kimberley Tan (MSc Thesis at Math Department) University of Oxford
 Formalisation of Synchronous Multiparty Session Types in Coq.
 supervisor: Nobuko Yoshida

ACADEMIC SERVICE

Journals	Journal of Symbolic Computation	Reviewer
Conferences	ITP, FSCD, LICS, CICM, ATVA ATVA (24 and 25)	Reviewer Member in Artifact Evaluation Committee

RECENT TALKS

Invited

08.03.2024	Decorated semantics for an imperative language with exceptions. Quantum Research Group, University of Oxford.
10.11.2023	SMTCoq – A Certified Checker for SMT. Mobility Reading Group, University of Oxford.
07.06.2019	Univalent Typoids in Coq . Ludwig-Maximilians-Universität, Munich-Germany.
27.12.2018	Decorated semantics for an imperative language with exceptions. Özyegin University, Istanbul-Turkey.
10.11.2015	Relative Hilbert-Post completeness for exceptions . Foundations Seminar, Ljubljana-Slovenia.

Others

- 10.09.2024 [Completeness of Asynchronous Session Tree Subtyping in Coq.](#)
[Fifteenth Conference on Interactive Theorem Proving.](#)
- 19.07.2024 [Completeness of Asynchronous Session Tree Subtyping in Coq.](#)
[Kent Concurrency Workshop and S-REPLS 15.](#)
- 01.02.2022 [Formal Categorical Reasoning.](#)
[Studies on Scientific Developments in Geometry, Algebra, and Applied Mathematics.](#)
- 27.03.2019 [Univalent Type Theory.](#)
Master Seminar, University of Innsbruck-Austria.
- 15.08.2018 [Concrete Semantics with Coq and CoqHammer.](#)
CICM'18 Hagenberg-Austria.
- 13.08.2018 [Towards Mac Lane's Comparison Theorem for the \(co\)Kleisli construction in Coq.](#)
FMM'18 Hagenberg-Austria.
- 31.05.2018 [SMTCoq, a plug-in for the trustworthy integration of SAT/SMT solvers into Coq.](#)
Master Seminar, University of Innsbruck-Austria.
- 03.06.2015 [IMP with exceptions over decorated logic.](#)
TFP'15, Sophia-Antipolis-France.
- 05.11.2014 [Program certification with computational effects.](#)
JNCF'14, Marseille-France.
- 11.07.2014 [Certification of programs with computational effects.](#)
CICM'14, Coimbra-Portugal.
- 09.01.2014 [Formal verification in Coq of program properties involving the global state effect.](#)
JFLA'2014, Fréjus - France.
- 05.12.2013 [Formal verification in Coq of program properties involving the global state effect.](#)
Séminaires LJK-Modèles et Algorithmes Déterministes: BIPOP-CASYS, Grenoble-France.

EVENTS ATTENDED

Invited

- 07.06.2019 [Types in Munich](#) (invited by Dr. Iosif Petrakis),
Ludwig-Maximilians-Universität, Munich-Germany.
- 09-13.11.15 University of Ljubljana (invited by Dr. Andrej Bauer),
Ljubljana, Slovenia.

Others

- 09-14.09.24 [Fifteenth Conference on Interactive Theorem Proving \(ITP'24\),](#)
Tbilisi-Georgia.
- 13-17.08.18 [Conferences on Intelligent Computer Mathematics \(CICM'18\),](#)
Hagenberg-Austria.
- 25-30.03.18 [3rd Conference on Artificial Intelligence and Theorem Proving \(AITP'18\),](#)
Aussois-France.
- 03-05.06.15 [The Sixteenth Symposium on Trends in Functional Programming \(TIFP'15\),](#)
INRIA, Sophia-Antipolis-France.

- 26-29.05.15 Colloque Inter'Actions en Mathématiques 2015 (organization committee),
Institut Fourier, Grenoble-France.
- 03-07.11.14 Journées Nationales de Calcul Formel (JNCF'14),
CIRM, Marseille-France.
- 07-11.07.14 Conferences on Intelligent Computer Mathematics (CICM'14),
Coimbra-Portugal.
- 07-18.04.14 PhD school on semantics of proofs and certified mathematics
Institut Henri Poincaré, CIRM, Marseille-France.
- 07-11.01.14 JFLA: Journées Francophones des Langages Applicatifs,
La-villa-clythia, Fréjus-France.
- 23-31.05.13 École Jeunes Chercheurs en Programmation,
Inria, Rennes-France.

CODE

Please see my [github repository](#).

MISCELLANEOUS

Programming	C/C++, Rust, Java, Haskell, OCaml, Coq, Lean.
Libraries	CRYMPIX, OpenMP, WHY3, SMTCoq, CoqHammer, Paco.

REFEREES

[Jean-Guillaume Dumas](#)

LJK, University Grenoble Alpes

jean-guillaume.dumas@univ-grenoble-alpes.fr

[Dominique Duval](#)

LJK, University Grenoble Alpes

dominique.duval@univ-grenoble-alpes.fr

[Cesare Tinelli](#)

Department of Computer Science,

The University of Iowa

cesare-tinelli@uiowa.edu

[Cezary Kaliszyk](#)

Computational Logic Group,

University of Innsbruck

cezary.kaliszyk@uibk.ac.at

[Murat Taylı](#)

Department of Computer Engineering,

Istanbul Kültür University

m.tayli@iku.edu.tr

[Nobuko Yoshida](#)

Department of Computer Science

University of Oxford

nobuko.yoshida@cs.ox.ac.uk